Pediatric Post-Resuscitation

PC-08 | Pediatric Cardiac

Quick flow

01 Sustained ROSC [EMT / AEMT / Paramedic]

Confirm pulse/perfusion and immediately stabilize airway, breathing, circulation, glucose, temperature, and rhythm.

02 Optimize Oxygenation / Ventilation [EMT / AEMT / Paramedic]

Once reliable SpO_2 is available, titrate toward 94-99% unless an individualized baseline applies. Avoid both hypocapnia and hypercapnia.

03 Support Perfusion [AEMT / Paramedic]

Maintain systolic and mean arterial pressure above the 10th percentile for age and sex. Avoid routine fluid boluses; use cautious reassessed fluid or Medical-Control-directed vasopressor support when indicated.

04 Temperature + Glucose [EMT / AEMT / Paramedic]

Check temperature and glucose. Avoid core temperature above 37.5 °C in the comatose child; correct hypoglycemia within standing orders.

05 Persistent Arrhythmia? [Paramedic]

Return to the rhythm-specific pediatric pathway. Continuing antiarrhythmic therapy requires Medical Control.

06 Pediatric-Capable Destination [EMT / AEMT / Paramedic]

Notify early, reassess continuously, and transport to the most appropriate pediatric-capable destination.

Overview

- For sustained ROSC in patients age 3 days through 15 years. Prevent recurrent arrest and secondary injury through controlled oxygenation, ventilation, perfusion, glucose, temperature, and rhythm management.
- Choose a destination capable of pediatric intensive care and the child's likely cardiac, neurologic, and temperature-management needs.

Provider-Level Actions

Use a length- or weight-based pediatric resuscitation system and perform immediate lifesaving care without delaying oxygenation, ventilation, CPR, or defibrillation.

EMT

- Maintain airway and appropriate ventilation, titrate oxygen toward SpO_2 94-99% when reliable unless a known individualized target applies, and monitor vital signs frequently.
- Check glucose when authorized, prevent fever, observe neurologic response, and reassess continuously for recurrent arrest.
- Use AO-02 Newly Born for patients through 3 days of age and the adult post-arrest pathway at age 16 years or older.

AEMT

- Perform all EMT care plus IV/IO access, glucose correction, cautious cause-directed fluid support when hypovolemia is suspected, and authorized post-airway analgesia/sedation.
- Do not administer a routine fluid bolus in post-arrest care; reassess perfusion and lungs after any fluid given.

Paramedic

- Perform all prior care plus cardiac/12-lead monitoring when it does not delay care, continuous waveform $EtCO_2$ after an advanced airway, rhythm-specific treatment, and Medical-Control-directed vasopressor support.
- Confirm advanced-airway position continuously with waveform capnography. Treat pain first, then provide age-appropriate sedation; paralysis alone is not sedation.
- Select a destination with pediatric ICU, cardiology, neurology, and temperature-management capability when available.

Indications

- Patient age 3 days through 15 years with sustained return of spontaneous circulation after respiratory or cardiac arrest.

Treatment steps

- Target normoxemia appropriate to the child's underlying condition. In most children with reliable pulse oximetry, titrate to SpO_2 94-99%; individualize for cyanotic congenital heart disease or another known baseline exception.
- Target ventilation appropriate to the child's underlying condition and limit exposure to hypocapnia and hypercapnia. Avoid excessive ventilation.
- Maintain both systolic and mean arterial blood pressure above the 10th percentile for age and sex; use the pediatric calculator and Medical Control for field targets.
- Avoid core temperature above 37.5 °C in the comatose child. Do not begin field targeted-temperature management unless specifically directed by Medical Control.
- If a dysrhythmia persists, return to its specific protocol. Continuing antiarrhythmic therapy requires Medical Control.

Warnings / precautions

- Do not use an adult oxygen target or force a usual saturation target in cyanotic congenital heart disease without individualized guidance.
- Do not administer routine fluid boluses or begin field targeted-temperature management without a specific indication and Medical Control direction.
- A paralyzed or ventilated child may be awake and in pain; treat pain before anxiety and maintain sedation.
- Post-ROSC condition can change rapidly; reassess continuously and return to the arrest pathway immediately if pulseless.

Clinical pearls

- Avoiding post-arrest fever and hypotension are key neuroprotective priorities.
- Pulse oximetry can overestimate oxygenation in patients with darker skin pigmentation; interpret the reading with the overall clinical condition.

Related pathways / contacts

AR-08 Post-Intubation / BIAD Management - Open for post-airway ventilation, analgesia, and monitoring.

PC-01 Pediatric Asystole / PEA - Open if nonshockable arrest recurs.

PC-04 Pediatric Pulseless Arrest - Open if the child becomes pulseless again.

PC-07 Pediatric VF / Pulseless VT - Open for recurrent shockable arrest.

Medication Reference - Open the current Claiborne medication reference.

Source and references

Claiborne County EMS approved protocol manual. | Current Tennessee EMS scope of practice and Claiborne County standing orders control for all provider permissions.

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